

ECE4010J Probabilistic Methods in Engineering

Extra Exercises and Solutions

Exercise 1 (10 pts) Let $X_1, X_2 \sim \text{Exponential}(\lambda)$ be iid random variables, with density

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0, \quad \lambda > 0.$$

Find the probability density function of

$$Y = \max(X_1, X_2).$$

Solution. First,

$$F_X(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}.$$

Then

$$\begin{aligned} F_Y(y) &= \Pr(Y \leq y) \\ &= \Pr(\max(X_1, X_2) \leq y) \\ &= \Pr(X_1 \leq y, X_2 \leq y) \\ &= F_X(y)F_X(y) \\ &= (1 - e^{-\lambda y})(1 - e^{-\lambda y}) \\ &= 1 - 2e^{-\lambda y} + e^{-2\lambda y}. \end{aligned}$$

Therefore,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = 2\lambda e^{-\lambda y} - 2\lambda e^{-2\lambda y}, \quad y > 0.$$

Hence

$$f_Y(y) = \begin{cases} 2\lambda e^{-\lambda y} - 2\lambda e^{-2\lambda y}, & y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Exercise 2 (10 pts) About 10% of light bulbs produced by a factory are defective. Use the normal distribution to approximate the probability that in a shipment of 150 light bulbs, between 15 and 20 inclusive are defective.

Explain what assumptions you use, the CDF of standard normal distribution. Make your estimate as accurate as possible.

Solution. Let X be the number of defective bulbs in the shipment. Then

$$X \sim \text{Binomial}(150, 0.1).$$

Since

$$\mu = np = 150(0.1) = 15, \quad \sigma^2 = np(1-p) = 150(0.1)(0.9) = 13.5,$$

we approximate

$$X \approx N(15, 13.5).$$

Using the continuity correction,

$$\begin{aligned}\Pr(15 \leq X \leq 20) &\approx \Phi\left(\frac{20.5 - 15}{\sqrt{13.5}}\right) - \Phi\left(\frac{14.5 - 15}{\sqrt{13.5}}\right) \\ &= \Phi(1.4969) - \Phi(-0.1361) \\ &\approx 0.4869.\end{aligned}$$

Exercise 3 (30 pts) A company receives 150 emails per day. On average, 100 of these emails are identified as spam. Assume that the number of spam emails received on each day follows a Poisson distribution. Recall that the Poisson distribution has pmf

$$\Pr(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad \lambda > 0.$$

- (i) **(10 pts)** What is the probability that a particular email is not spam?

Solution. The average spam rate is

$$\lambda = \frac{100}{150} = \frac{2}{3}.$$

Let X denote the number of spam classifications for one particular email. Then

$$X \sim \text{Poisson}\left(\lambda = \frac{2}{3}\right), \quad \Pr(X = k) = \frac{e^{-2/3} \left(\frac{2}{3}\right)^k}{k!}.$$

Thus

$$\Pr(X = 0) = e^{-2/3} = 0.513417.$$

- (ii) **(10 pts)** What is the probability that a particular email is classified as spam at least 2 times, e.g. flagged multiple times due to multiple filters triggering independently?

Solution.

$$\begin{aligned}\Pr(X \geq 2) &= 1 - \Pr(X = 0) - \Pr(X = 1) \\ &= 1 - e^{-2/3} - \frac{2}{3}e^{-2/3} \\ &= 0.144305.\end{aligned}$$

- (iii) **(10 pts)** What is the expected number of spam emails in any randomly selected group of 10 emails?

Solution. Here $\lambda = \frac{2}{3}$ and $S = 10$, so

$$\lambda S = \frac{20}{3} =: k.$$

Let Y denote the number of spam emails in 10 emails. Then

$$Y \sim \text{Poisson}\left(\frac{20}{3}\right),$$

and hence

$$E[Y] = k = \frac{20}{3}.$$

Exercise 4 (30 pts) Let X and Y be random variables with joint density

$$f_{X,Y}(x, y) = \begin{cases} \frac{y(y+1)e^{-y}}{x^{y+2}}, & 1 \leq x < \infty, \quad 0 < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

(i) **(7 pts)** Calculate the marginal density of Y .

Solution.

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \\ &= y(y+1)e^{-y} \int_1^{\infty} x^{-(y+2)} dx \\ &= y(y+1)e^{-y} \left[-\frac{1}{y+1} x^{-(y+1)} \right]_1^{\infty} \\ &= ye^{-y}. \end{aligned}$$

Therefore,

$$f_Y(y) = \begin{cases} ye^{-y}, & 0 < y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

(ii) **(7 pts)** Calculate the conditional density of $X \mid Y = y$ for fixed y .

Solution.

$$\begin{aligned} f_{X|Y}(x \mid y) &= \frac{f_{X,Y}(x, y)}{f_Y(y)} \\ &= \frac{\frac{y(y+1)e^{-y}}{x^{y+2}}}{ye^{-y}} \\ &= \frac{y+1}{x^{y+2}}. \end{aligned}$$

Thus, for fixed $y > 0$,

$$f_{X|Y}(x \mid y) = \begin{cases} \frac{y+1}{x^{y+2}}, & 1 \leq x < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

(iii) **(7 pts)** Calculate $E[X \mid Y]$.

Solution.

$$\begin{aligned} E[X | Y = y] &= \int_1^{\infty} x f_{X|Y}(x | y) dx \\ &= \int_1^{\infty} x \frac{y+1}{x^{y+2}} dx \\ &= (y+1) \int_1^{\infty} x^{-(y+1)} dx \\ &= (y+1) \left[-\frac{1}{y} x^{-y} \right]_1^{\infty} \\ &= \frac{y+1}{y} \\ &= 1 + \frac{1}{y}. \end{aligned}$$

Therefore,

$$E[X | Y] = 1 + \frac{1}{Y}, \quad 0 < Y < \infty.$$

(iv) **(7 pts)** Calculate $E[X]$.

Solution.

$$\begin{aligned} E[X] &= E[E[X | Y]] \\ &= \int_0^{\infty} E[X | Y = y] f_Y(y) dy \\ &= \int_0^{\infty} \left(1 + \frac{1}{y} \right) y e^{-y} dy \\ &= \int_0^{\infty} (y e^{-y} + e^{-y}) dy \\ &= \int_0^{\infty} y e^{-y} dy + 1 \\ &= 2. \end{aligned}$$

(v) **(2 pts)** Are X and Y independent? Explain.

Solution. No. If X and Y are independent, then

$$E[X | Y] = E[X].$$

However,

$$E[X | Y] = 1 + \frac{1}{Y}, \quad E[X] = 2,$$

so $E[X | Y] \neq E[X]$ in general. Therefore, X and Y are not independent.

Exercise 5 (20 pts) Given function $g : \mathbb{R} \rightarrow \mathbb{R}$ as

$$g(x) = \begin{cases} 1, & -2 \leq x < -1, \\ x^2, & -1 \leq x < 0, \\ x, & 0 \leq x < 2, \\ 2, & 2 \leq x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

If $Y = g(X)$ and $X \sim \text{Uniform}([-4, 4])$, i.e. uniform distribution over $[-4, 4]$, find the CDF of Y .

Solution. Since $X \sim \text{Uniform}([-4, 4])$,

$$f_X(x) = \frac{1}{8},$$

with

$$F_X(x) = \begin{cases} 0, & x \leq -4, \\ \frac{1}{8}x + \frac{1}{2}, & -4 < x < 4, \\ 1, & x \geq 4. \end{cases}$$

We compute

$$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y).$$

1) If $y < 0$, then

$$F_Y(y) = \Pr(g(X) \leq 0) = 0.$$

2) If $0 \leq y < 1$, then

$$\begin{aligned} F_Y(y) &= \Pr(-\sqrt{y} \leq X \leq y \cup X \geq 3 \cup X < -2) \\ &= F_X(y) - F_X(-\sqrt{y}) + 1 - F_X(3) + F_X(-2) \\ &= \frac{1}{8}y + \frac{1}{8}\sqrt{y} + \frac{3}{8}. \end{aligned}$$

3) If $1 \leq y < 2$, then

$$\begin{aligned} F_Y(y) &= \Pr(X \leq y \cup X \geq 3) \\ &= F_X(y) + 1 - F_X(3) \\ &= \frac{1}{8}y + \frac{5}{8}. \end{aligned}$$

4) If $y \geq 2$, then

$$F_Y(y) = \Pr(g(X) \leq y) = 1.$$

Hence the CDF is

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{1}{8}y + \frac{1}{8}\sqrt{y} + \frac{3}{8}, & 0 \leq y < 1, \\ \frac{1}{8}y + \frac{5}{8}, & 1 \leq y < 2, \\ 1, & y \geq 2. \end{cases}$$